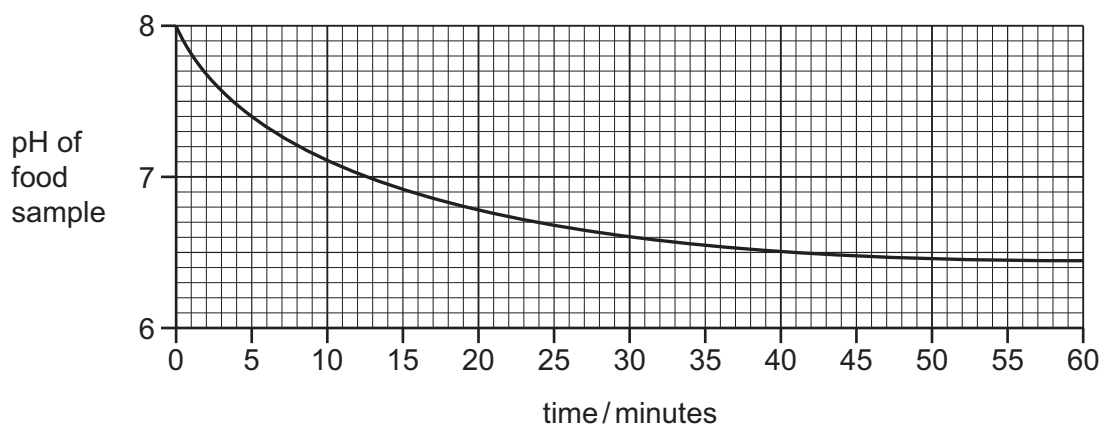


- 13** Lipase is an enzyme that catalyses the hydrolysis of lipids. An experiment was carried out to investigate changes in pH when lipase is added to a food sample containing a high proportion of lipids.

The results are shown in the graph.



Which statements are possible explanations of the results of the experiment between 50 minutes and 60 minutes?

- 1 Enzyme concentration becomes the limiting factor.
- 2 Substrate concentration becomes the limiting factor.
- 3 All the enzyme active sites are occupied.
- 4 Denaturation of the enzyme by the products takes place.
- 5 Products are acting as competitive inhibitors.

A 1, 2 and 3 **B** 1, 4 and 5 **C** 2, 3 and 4 **D** 2, 4 and 5